

LLM: Architecture Variants

Yaxshi model chiqishi uchun hozrgi frontier modellar quyidagi modelni **hususiyatlarini** kuchaytirishadi:

- tezlik
- aqilli
- less memory usage
- uzun context tushunish

3 ta architecture variantlar bor:

- **MHA** (multi-head attention)
- **MQA** (multi-query attention)
- **GQA** (group-query attention)

Eski modellarda misol **GPT-2 da MHA** architecture variant ishlatilingan. Va **12 head** bulgan. U degani, **parallel** holda **12 attention operation** buladi.

Component	GPT-2	Frontier models
Attention	MHA	GQA, MQA
Position	Absolute embedding	RoPe (rotary position)
Activation	GELU	SwiGLU
Normalization	LayerNorm	RMSNorm
Vocabulary	BPE (~50K tokens)	Larger BPE, SentencePiece
Goal	Good LM	Fast answer, high quality

Residual: layerlar juda kup buladi. va bitta layer ishini bajarib keyingisiga utyotkanda malum bir malumotlar yuqolib qoladi. Shuni oldini olish uchun residual connection qilamiz